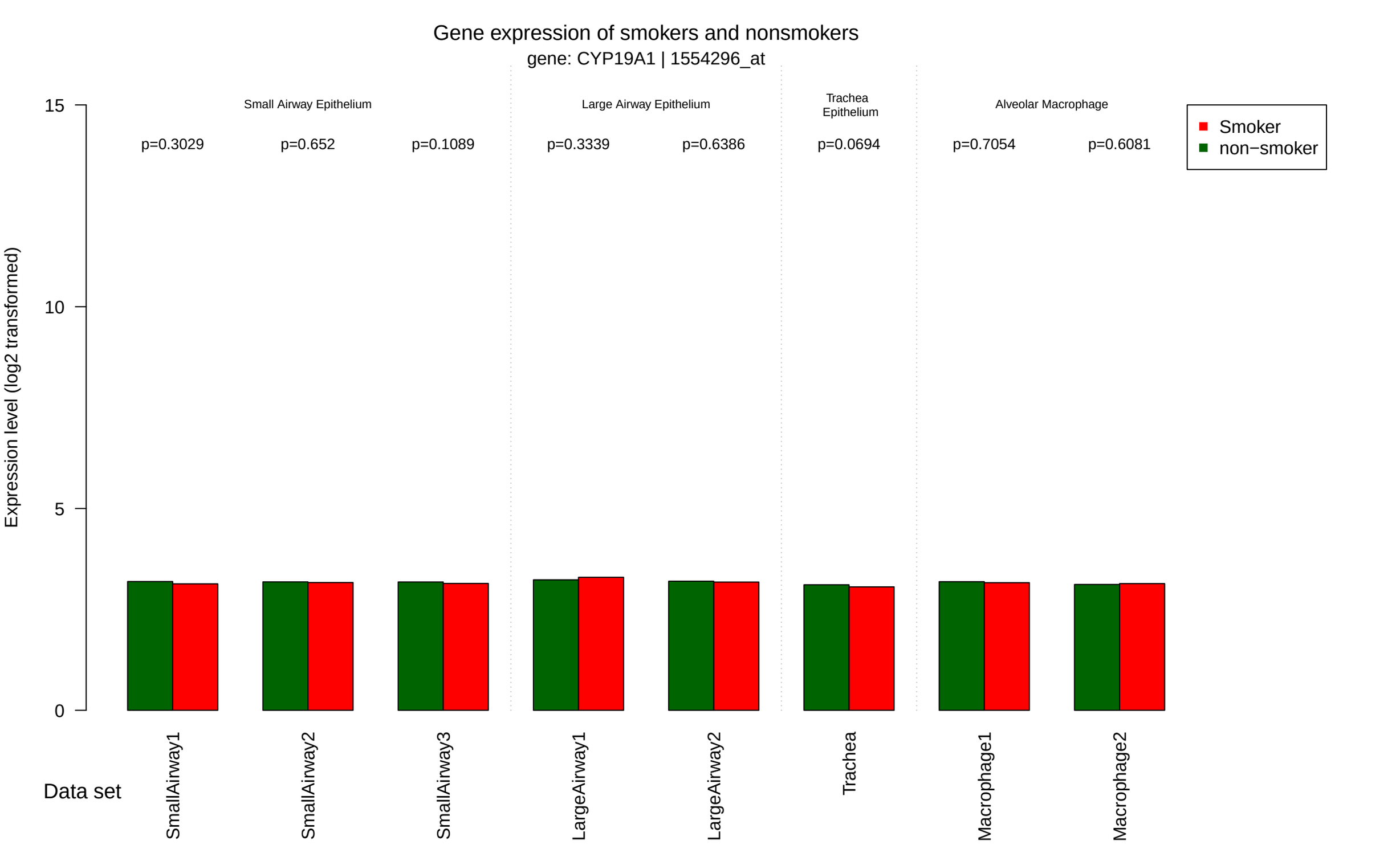
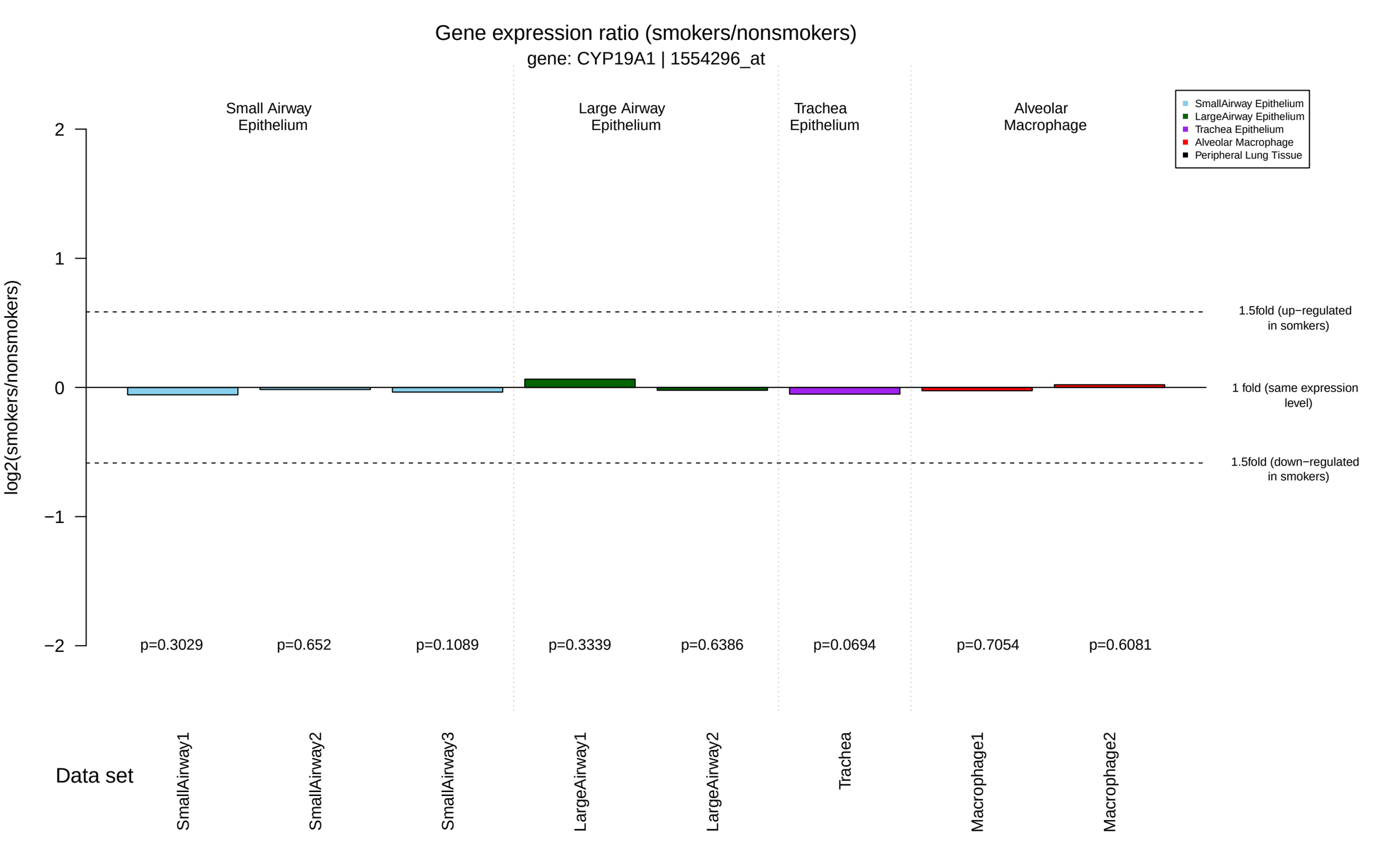
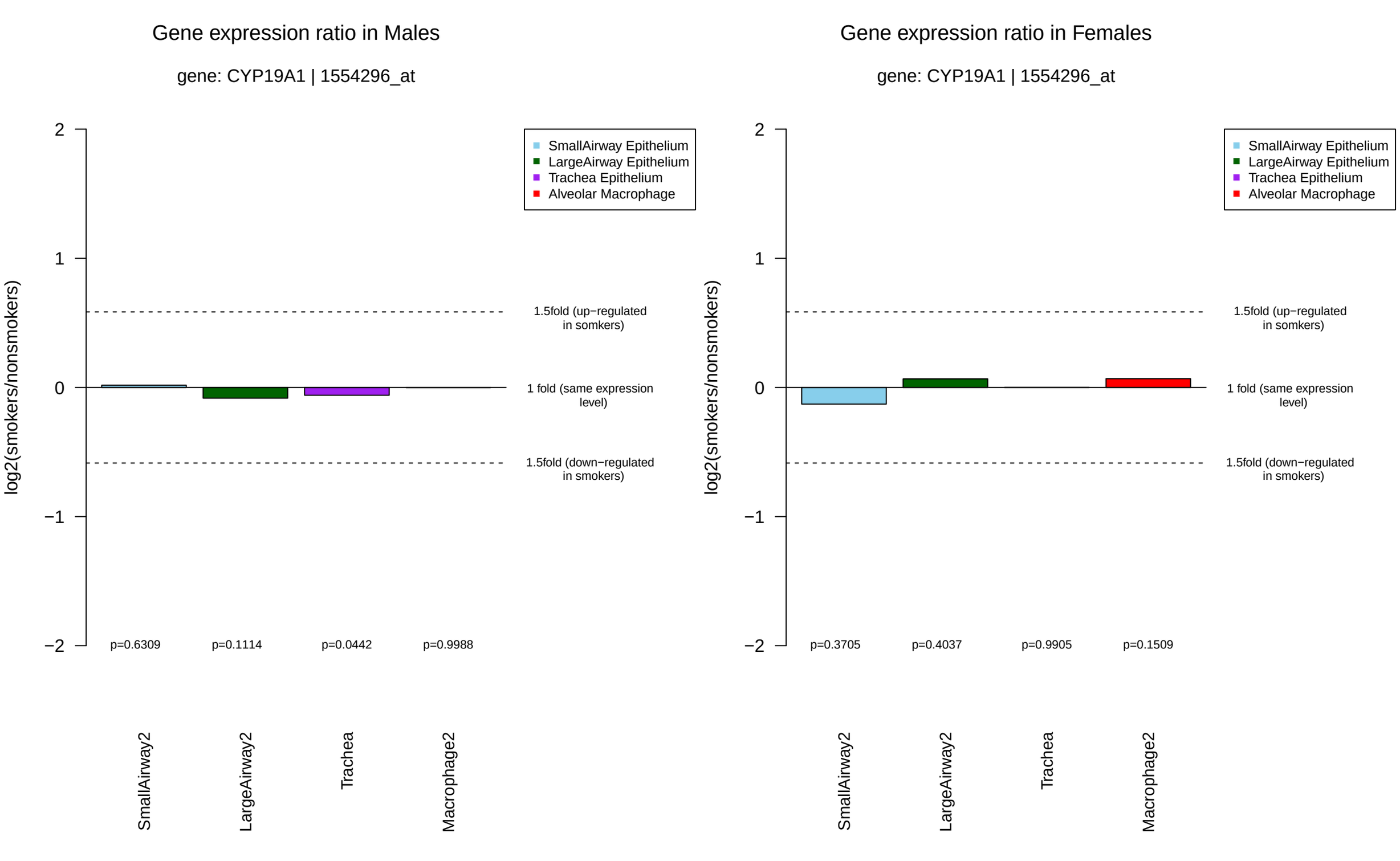
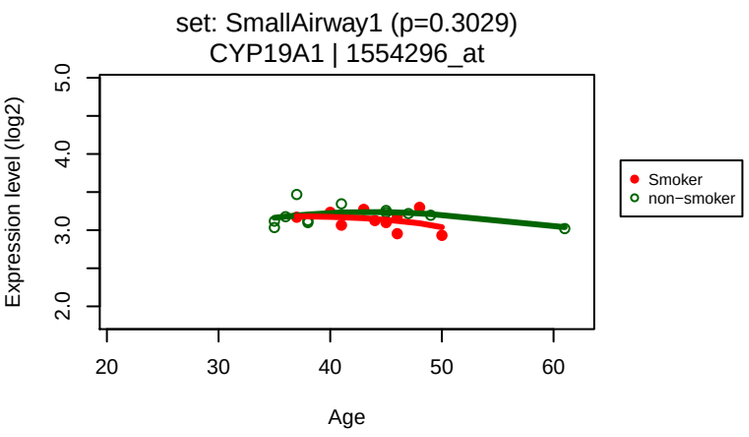
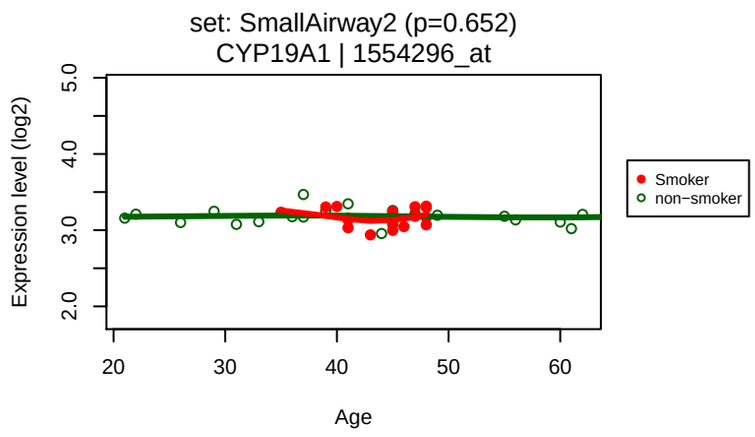
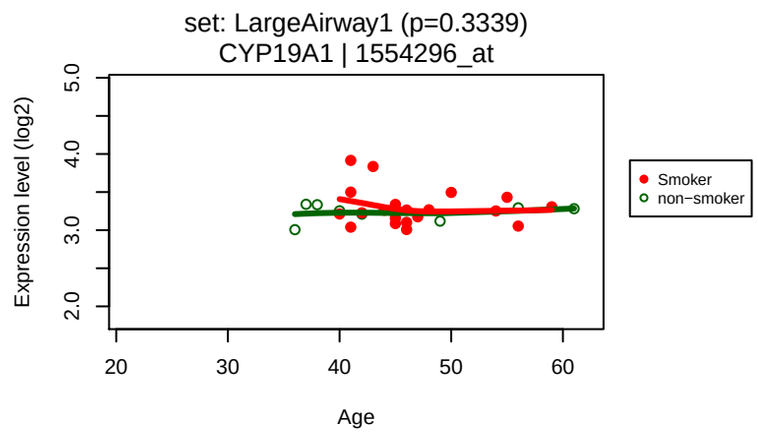

Gene expression of smokers and nonsmokers
gene: CYP19A1 | 1554296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3029
p=0.652
p=0.1089
p=0.3339
p=0.6386
p=0.0694
p=0.7054
p=0.6081
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

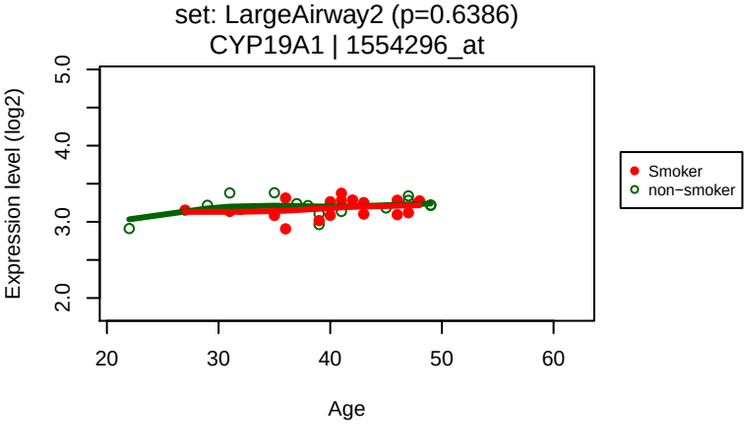

Gene expression ratio (smokers/nonsmokers)
gene: CYP19A1 | 1554296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3029
p=0.652
p=0.1089
p=0.3339
p=0.6386
p=0.0694
p=0.7054
p=0.6081
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

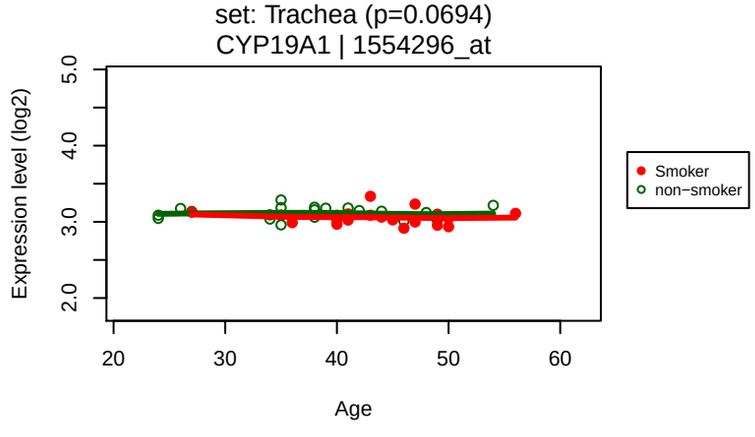

Gene expression ratio in Males
gene: CYP19A1 | 1554296_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6309
p=0.1114
p=0.0442
p=0.9988
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP19A1 | 1554296_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3705
p=0.4037
p=0.9905
p=0.1509
SmallAirway2
LargeAirway2
Trachea
Macrophage2

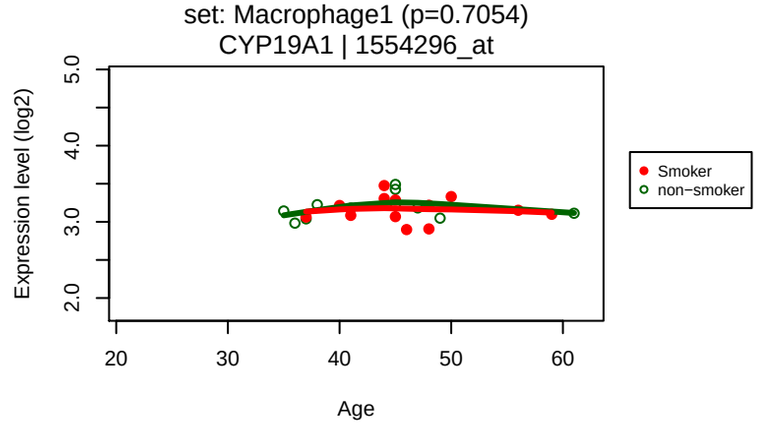

set: SmallAirway1 (p=0.3029)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker

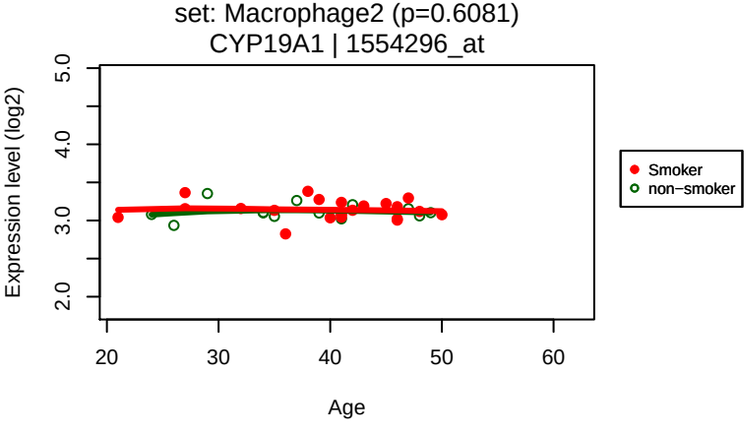

set: SmallAirway2 (p=0.652)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker

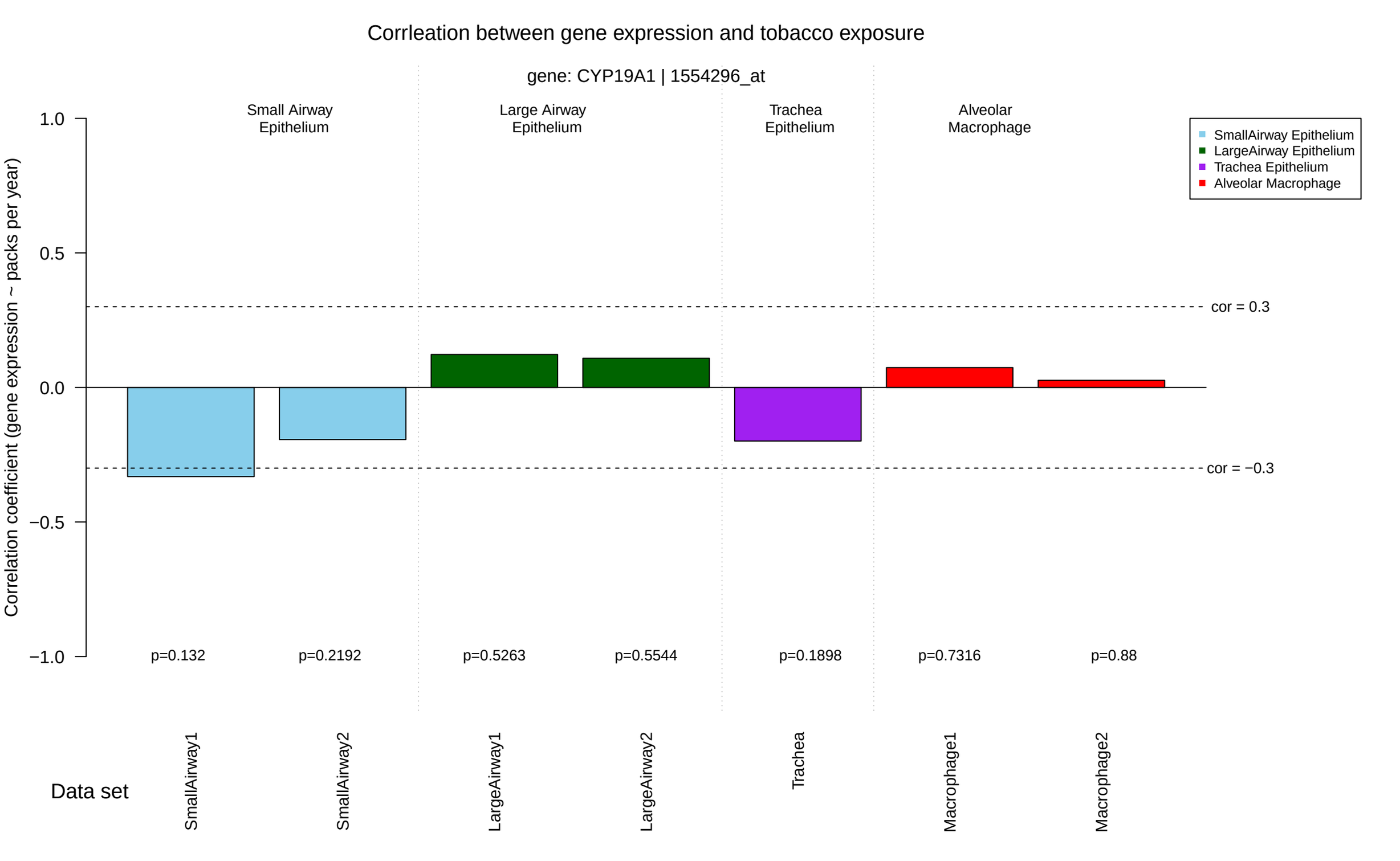

set: LargeAirway1 (p=0.3339)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6386)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0694)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7054)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6081)
CYP19A1 | 1554296_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP19A1 | 1554296_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.132
p=0.2192
p=0.5263
p=0.5544
p=0.1898
p=0.7316
p=0.88
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2